Cyclistic Bike-Share Case Study Analysis Report

1. Introduction

Cyclistic is a bike-share company that offers flexible transportation services through single-ride passes and annual memberships. While casual riders generate revenue, annual members provide more consistent and predictable income. Therefore, **the company aims to increase the number of annual memberships.**

This analysis explores the behavioural differences between casual riders and members to support data-driven marketing strategies that encourage conversion from casual to member users.

2. Business Task

The primary business question guiding this analysis is:

How do casual riders and annual members use Cyclistic bikes differently?

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3. Data Source

Cyclistic Bike-Share Dataset

Time period: Jan 2025 – Dec 2025

You can download the dataset from this ***link***

Tools Used for Analysis

- Excel
- Power Query
- Pivot Tables
- Data Visualization

4. Data Cleaning and Preparation

Data cleaning and transformation were conducted to ensure accuracy and usability:

- Combined 12 monthly datasets using Power Query
- Removed duplicates and irrelevant columns containing only null values
- Replaced missing station values with “Unknown”
- Converted date-time fields into proper formats using locale settings
- Created new derived columns:
    - Day of the week
    - Month of the year
    - Hour of the day
    - Time of day
    - Ride length

- Verified and corrected data types (date, time, numeric, text)

These steps ensured a clean and structured dataset suitable for analysis.

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5. Methodology

The analysis was conducted using Pivot Tables and charts in Excel. A comparative approach was applied across all key variables by segmenting data into:

- Member riders
- Casual riders

The following key metrics were analysed:

- Average ride length per week
- Ride frequency by day of the week
- Ride frequency by hour/time of the day
- Ride frequency by month of the year

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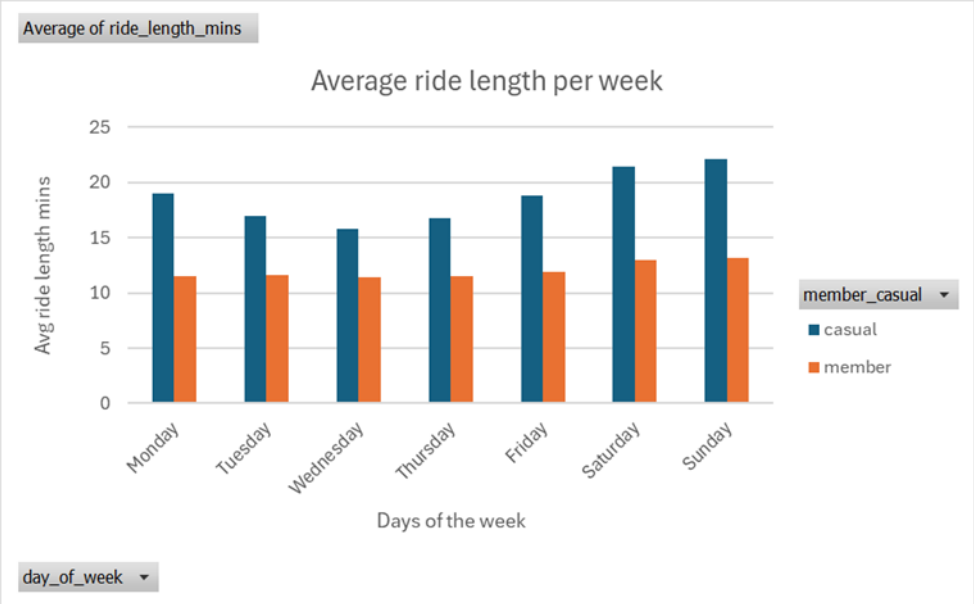
6. Analysis and Findings

1. Average Ride Length per week



- Casual riders have a significantly higher average ride duration compared to member.
  Casual riders tend to use the service for longer, leisure-oriented trips, while members use it for shorter, routine journeys.

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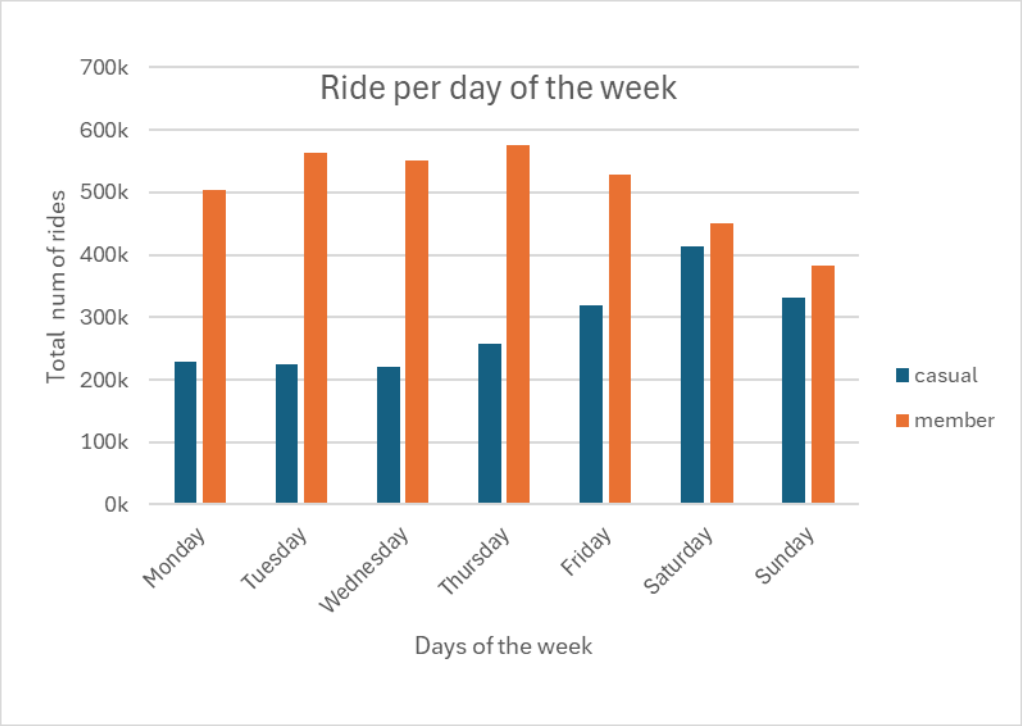
2. Weekly ride patterns: Member vs Casual



- Casual riders show increased activity on weekends, while members maintain relatively consistent usage throughout the week.
- Members primarily use the service for weekday commuting, whereas casual riders prefer weekend recreational use.

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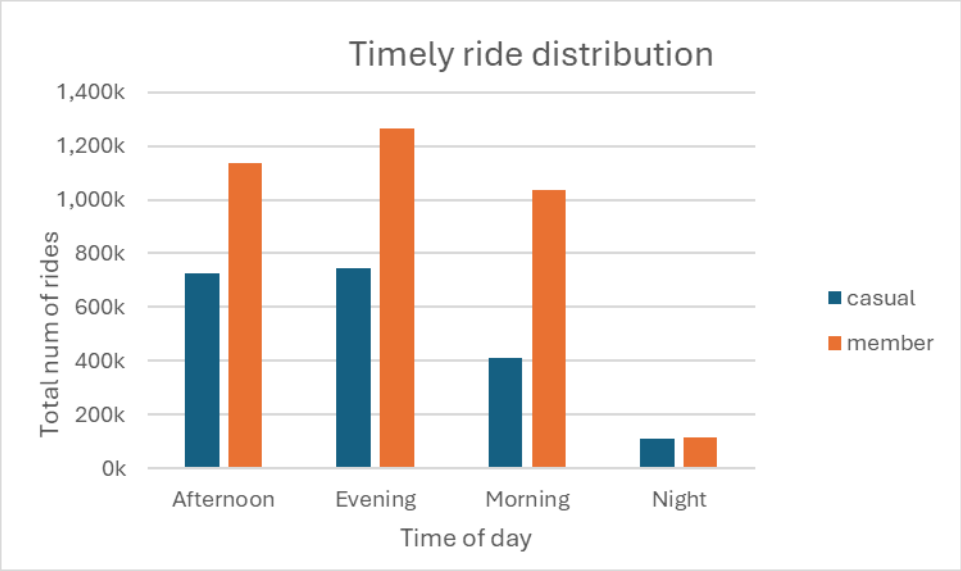
3. Ride distribution by Time of day

- **Afternoon** (12–4:59 pm) had the highest number of riders, which were mainly the member users. **Evening** (5- 8 pm) had the highest number of rides with mainly member users, then the **Morning** (6–11:59 am) with mainly member users, lastly **Night** (8:01 pm-5;39am) with her percent of casual users.



- Members exhibit clear peak usage during morning and Evening commuting hours, while casual riders show more evenly distributed usage throughout the day.
- This pattern reinforces that members rely on the service for commuting, while casual riders use it more flexibly.

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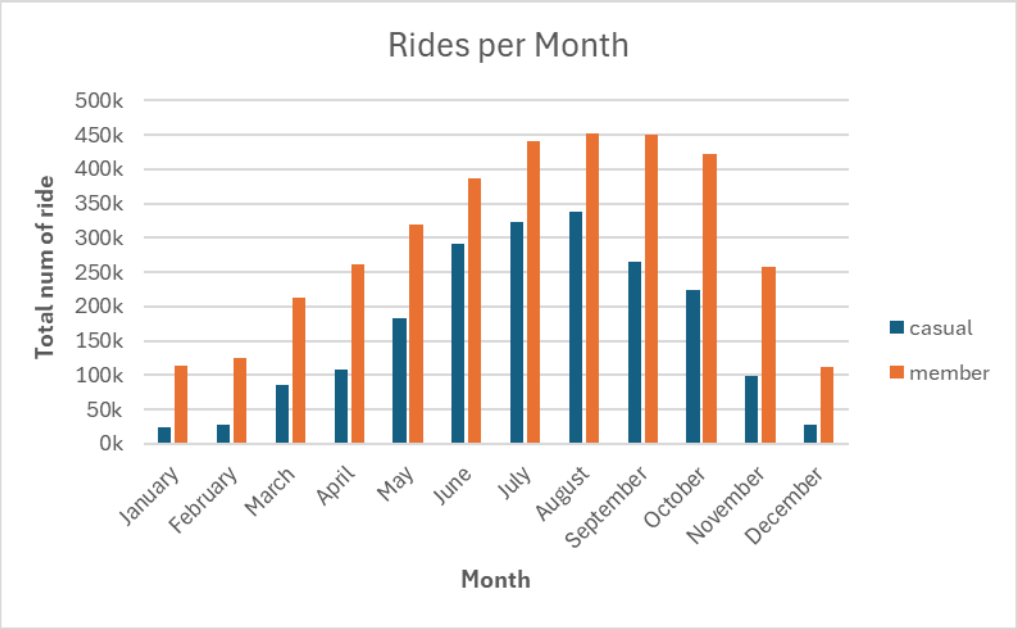
4. Rides per Month



- Both user types show increased activity during summer months, indicating seasonality.

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7. Recommendations

Based on the findings, the following recommendations are proposed:

- **Convert Casual Riders to Members**

Introduce targeted promotions such as free trials, discounts, or limited-time membership offers, especially during weekends when casual usage is highest.

- **Introduce Flexible Pricing Plans**

Offer pricing options tailored to casual riders, such as hourly packages or day passes, to better match their usage patterns.

- **Focus Marketing on Leisure Segments**

Develop campaigns targeting tourists and weekend users, emphasising the convenience and enjoyment of bike-sharing.

- **Enhance User Experience**

Simplify the onboarding process and improve app usability to encourage repeat usage and conversion to membership.

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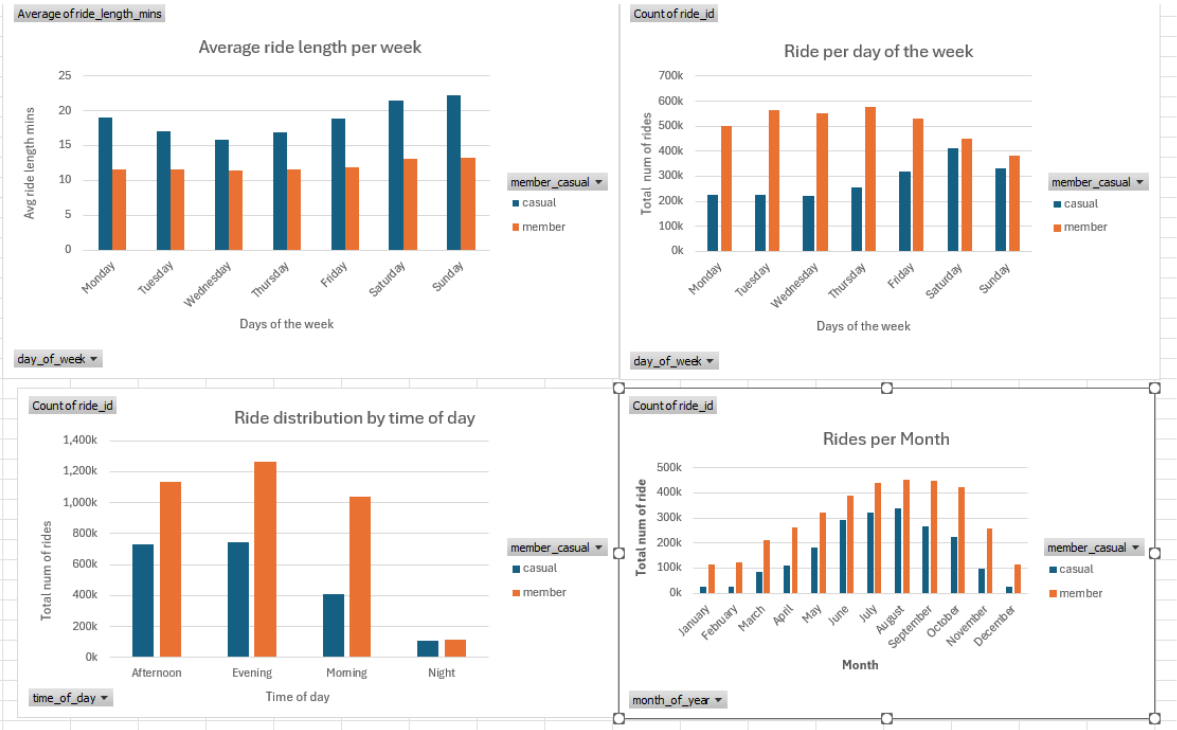
8. Dashboard Visualization